Road Accident Risk Predictor

Explainable Risk Scoring for Infrastructure Safety Prioritization

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Summary

Machine learning identifies which road conditions carry disproportionate accident risk, so limited infrastructure safety spending can be targeted rather than spread evenly — with SHAP explanations for every prediction.

Problem

Road safety agencies cannot treat every road segment equally with finite inspection and engineering resources. Road trauma is concentrated in particular conditions rather than spread evenly across a network. The challenge is identifying which conditions carry disproportionate risk so intervention — signage, lighting, speed-limit review — can be prioritized where it has the most potential benefit.

The scale of that problem in New Zealand is documented. A separate national analysis by the same author, of the NZTA Waka Kotahi Crash Analysis System, covers 153,597 reported injury crashes between 2010 and 2024, of which 21.43% killed or seriously injured someone, and finds the speed environment strongly associated with severity: roads posted at 90 km/h or above carry an odds ratio of 1.835 [1.787, 1.884] against roads at 50 km/h or below.

That analysis also establishes the limitation that motivates this tool. It models severity **given that a crash has already occurred**. It cannot indicate where crashes will occur, because the public extract excludes non-injury crashes by design and carries no exposure denominator. Prioritising limited engineering budget requires the other question — which conditions carry elevated risk of a crash in the first place — and that is the question this project addresses.

Relationship to that companion research

These are two separate pieces of work on the same problem, deliberately kept distinct. The CAS analysis uses real New Zealand crash records to characterise **severity**. This tool is trained on an open synthetic competition dataset to score condition-based **risk**. The CAS results do **not** validate this model and are not used to do so. They establish two things only: that condition-based reasoning about road harm is worth doing in the New Zealand context, and that severity data alone cannot support prioritisation.

The distinction matters in a way worth stating plainly, because the two can point in opposite directions on the same variable. The CAS analysis finds precipitation associated with *lower* odds of serious injury, consistent with behavioural compensation, while risk models of this kind generally treat adverse weather as risk-increasing. Both can hold at once: rain may reduce the severity of crashes that happen while increasing how many happen. That is precisely why crash frequency and crash severity should not be inferred from one another, and why a prioritisation tool cannot be built from severity data alone.

Solution

A Random Forest Regressor trained on 517,754 road-condition scenarios (15 features, including three engineered interaction terms) predicts a risk score for a given road/environment combination and explains *why* via SHAP feature attribution — not just a number, but which factors are driving it (speed, curvature, lighting, weather). Live and publicly deployed at the link above.

Evidence of impact

Ranking all training rows by predicted risk: the **top 10% of road conditions carry 1.93× their proportional share of total predicted risk** (19.3% of total risk mass concentrated in 10% of segments). This is measured directly from the model's own output, not assumed.

Risk tier	% of segments	Share of total risk mass	Concentration ratio
Top 1%	1%	2.5%	2.50×
Top 5%	5%	11.1%	2.22×
Top 10%	10%	19.3%	1.93×
Top 25%	25%	41.0%	1.64×

Under scenario-based intervention-effectiveness assumptions (15–30%, sourced from FHWA's Crash Modification Factors Clearinghouse — a repository of 3,000+ peer-reviewed countermeasure studies), this represents a targeting mechanism with measurable potential benefit. **This is stated as a scenario-based potential impact, not a causal claim** — the model has not been shown to have prevented any crash. Full methodology and worked calculation: `IMPACT.md` in the supporting files.

SDG alignment

Primary — SDG 3.6: halve global deaths and injuries from road traffic accidents.
Secondary — SDG 11.2: safe, affordable, sustainable transport systems.

Mechanism: risk scoring → targeted infrastructure spend → fewer serious injuries. The model does not predict who will crash — it identifies which road conditions carry disproportionate predicted risk.

What is innovative

1. **Predictive** — moves from historical reporting toward forward-looking risk prioritization.
2. **Explainable** — SHAP translates every prediction into interpretable contributing factors, not a black box.
3. **Action-oriented** — connects risk scores to infrastructure intervention prioritization, not just a number on a screen.

Technical evidence

Random Forest Regressor, 200 trees, 517,754 training rows, validation R^2 0.88 — independently reproduced (0.8835), not just claimed. 5-fold CV std ±0.0003. No data leakage: no duplicate IDs, and the dataset has no geographic or temporal columns for a random split to leak across. Subgroup performance checked by road type and weather (stable, R^2 0.877–0.885) and by lighting (small honest degradation toward night, R^2 0.857→0.846 — disclosed, not hidden). Full detail: `MODEL_VALIDATION.md`.

Responsible AI

The model uses road and environmental conditions only — no driver identity, ethnicity, or demographic data. Intended for infrastructure planning and safety prioritization, explicitly **not** for individual driver liability, profiling, or enforcement. This is a live decision- support prototype, not a production system with an operator or SLA. Known limitations, including a disclosed subgroup-performance gap in low-light conditions and the boundary of what this synthetic (Kaggle Playground Series S5E10) dataset can support, are documented in full in `RESPONSIBLE_AI.md`.

Live demo

https://roadaccident-roshanar-aryal.streamlit.app/

Hosted on Streamlit Community Cloud’s free tier, which sleeps after inactivity — cold start measured at ~95 seconds. If the link shows “Zzzz... this app has gone to sleep,” click “Yes, get this app back up!” and wait roughly a minute. Screenshots and a walkthrough recording are included in the supporting files as a fallback.

Contents of the supporting zip

- This document (`PROJECT_SUBMISSION.pdf`)
- `screenshots/` — live gauge prediction with SHAP contributions, model info and feature importance

- `walkthrough.mp4` — 60–90s recorded demo (if included)
- `IMPACT.md`, `RESPONSIBLE_AI.md`, `MODEL_VALIDATION.md` — full supporting detail behind the summaries above